Monetary Policy as a Boundary Condition in the IRFR Framework

A restructured draft note connecting policy intervention, boundary-value problems, repeated-root volatility, and Ritchken–Chuang-style humped volatility

Draft working note

Abstract

Monetary policy operates at the short end of the interest-rate curve, directly influencing the overnight rate while leaving the remainder of the forward curve to market determination. This paper argues that the appropriate mathematical representation of this asymmetry is through boundary conditions imposed at zero maturity. The paper is a companion extension to the core IRFR working paper of Oza [1], which develops the underlying single-factor term-structure model.

Within the instantaneous rolling forward rate (IRFR) framework, we show that these boundary conditions do not merely affect local behaviour at the short end, but instead determine the admissible functional form of forward-rate volatility across maturities. In particular, when the maturity-decay and calendar-time drift-diffusion scales coincide, corresponding to the branch $r = 1$, the boundary-value problem becomes a coalescing-root system whose independent solution includes the mode

$$\tau e^{-a^2\tau}.$$

This extension keeps the first three IRFR axioms unchanged: stochastic evolution, rolling-maturity invariance, and equilibrium moment transport remain the structural machinery of the model. What changes is the short-end boundary closure. In the original non-intervention benchmark, Axiom 4 imposes the natural positivity boundary at zero maturity. In the present paper, monetary policy is represented as a change in that boundary closure rather than as a change in the underlying rolling-maturity logic.

The fixed-boundary, or Dirichlet, case is not proposed as a literal description of monetary policy. It is a clean limiting benchmark: the short rate is pinned at a policy value, so short-end variance must vanish and risk is displaced into the interior of the curve. The more realistic policy case is Robin: the short end is only partially anchored, as policy attempts to moderate short-term rate moves in pursuit of price-stability and growth objectives.

As a result, the commonly observed volatility hump emerges as a boundary-induced eigenmode rather than a modelling assumption. Throughout, the drift-diffusion scales a_N, a_D, and a_R are kept distinct for their respective boundary conditions, and no cross-regime equality of a, MSNR, or total variance is assumed without separate justification.

1 Motivation: policy intervention as a boundary-value problem

The starting economic observation is deliberately simple: policy intervention is concentrated at the short end. Central banks set or steer policy rates, reserve remuneration, and operating targets. They do not mechanically set the thirty-year forward rate. Longer maturities remain market objects, shaped by expectations, credibility, term premia, and the anticipated path of future policy.

This gives the paper its main interpretation. Term-structure volatility is not treated as a freely chosen curve shape. Instead, volatility shapes are generated by the boundary conditions imposed on the forward-rate domain. The volatility hump is therefore a boundary-induced eigenmode: it appears because the policy boundary suppresses, or partially suppresses, short-end stochasticity and forces the remaining stochastic exposure to propagate into the interior of the maturity curve.

This makes the language of boundary conditions natural. In a physical boundary-value problem, one does not need to control the whole domain to change the solution. It is enough to change what is allowed at the boundary. A fixed endpoint, a free endpoint, and a partially constrained endpoint produce different shapes inside the domain. The same idea can be applied to the IRFR curve: a policy regime changes the short-end condition that the forward curve must satisfy.

This point also clarifies the role of the original positivity condition. In the non-intervention case, the relevant short-end closure is the natural lower boundary: as the short-maturity IRFR approaches zero, the stochastic loading vanishes and positivity is preserved. A policy intervention changes the economic boundary being enforced. A fully pinned policy rate is the Dirichlet limiting case; a central bank that leans against short-rate movements without perfectly fixing them is naturally represented by a Robin condition. Thus this paper treats Axiom 4 not as abandoned, but as generalized into a boundary-regime condition.

The Dirichlet case is useful precisely because it is too strong. It represents the limiting case in which the short rate is pinned at a policy value x_D. This is not a realistic description of day-to-day monetary policy, but it provides a clean benchmark. If the short end is fixed, then stochastic variation cannot enter at zero maturity. Any remaining uncertainty must be pushed away from the boundary into the interior of the curve. That displacement is the intuition behind the volatility hump.

The Robin case is the more realistic policy-intervention construct. In practice, policy does not pin the short end perfectly; it leans against short-rate moves. That is consistent with policy mandates that seek to moderate inflation, support growth, and stabilise financial conditions. A Robin boundary captures this by linking the short-end slope of the curve to the deviation of the short rate from the policy boundary. It is neither a fully free market boundary nor a perfectly fixed policy boundary.

The intended policy interpretation is therefore:

- **Neumann or free boundary:** the short end is market-determined and remains stochastic.
- **Dirichlet or fixed boundary:** the short end is pinned at a policy value x_D. This is a theoretical limiting case and a building block.
- **Robin or mixed boundary:** the short end is partially anchored to x_D. This is the preferred policy-intervention representation.

The paper is built in four stages. First, it uses the general r-branch machinery to define the Dirichlet boundary problem for $0 < r_D \leq 1$. Second, it explains why $r_D = 1$ is not a naive substitution into that distinct-root formula, but a coalescing-root case in which the independent boundary mode becomes $\tau e^{-a_D^2 \tau}$. Third, it compares the repeated-root Neumann and Dirichlet benchmarks side by side, showing how the fixed boundary replaces the ordinary exponential stochastic mode with the repeated-root boundary mode. Fourth, it uses that repeated-root structure to develop the Robin boundary condition and connect the framework to empirical humped-volatility specifications.

2 Reference IRFR machinery from the working paper

The working paper derives affine drift and diffusion forms from rolling-maturity invariance and affine closure. It defines the ratio

$$r := \frac{\lambda^2}{a^2}, \tag{2.1}$$

which separates the maturity-decay scale λ^2 from the temporal drift-diffusion scale a^2. The working paper also establishes that, within a given structural regime, the drift and diffusion share the same maturity loading. Consequently, the local market signal-to-noise ratio is

$$\mathrm{MSNR} = \frac{a}{\sqrt{2}}. \tag{2.2}$$

A key discipline in the boundary-condition paper is that the drift-diffusion scale is regime-indexed. We therefore write

$$a_N, \qquad a_D, \qquad a_R \tag{2.3}$$

for Neumann, Dirichlet, and Robin regimes respectively. Cross-regime equality is not assumed in this paper.

2.1 Notation convention for drift, diffusion, loading, and variance rate

Throughout this note, the notation follows the IRFR convention in which f denotes the drift-side coefficient in the Kolmogorov forward equation and g denotes the half-volatility coefficient. Thus the Itô volatility coefficient is

$$\sigma_B(\tau) = \sqrt{2}\, g_B(\tau), \tag{2.4}$$

for a boundary regime $B \in \{N, D, R\}$. When the regime admits a maturity loading $H_B(\tau)$, we write

$$g_B(\tau) = a_B H_B(\tau), \qquad \sigma_B(\tau) = \sqrt{2}\, a_B |H_B(\tau)|. \tag{2.5}$$

The corresponding local instantaneous variance rate is

$$v_B(\tau) := \frac{\partial}{\partial t}\,\mathrm{Var}(x^B_{t,t+\tau}) = 2 g_B(\tau)^2 = 2a_B^2 H_B(\tau)^2. \tag{2.6}$$

This convention separates three objects that should not be conflated: g_B is the half-volatility coefficient, σ_B is the Itô volatility coefficient, and H_B is the maturity loading. The paper uses v_B for the pointwise variance rate and reserves total variance for the rank-one integrated-loading object.

2.2 Axioms 1–3 and the boundary closure condition

The boundary-condition extension does not change the first three axioms of the IRFR framework. Axiom 1 supplies the diffusion setting; Axiom 2 imposes rolling-maturity invariance; and Axiom 3 governs equilibrium moment transport. These are the structural restrictions that make the model an IRFR model.

The point of departure is Axiom 4. In the original paper, Axiom 4 is stated as the positivity boundary at zero maturity: the short-maturity IRFR remains positive and the diffusion coefficient vanishes as the short-maturity rate approaches zero. That condition is appropriate for the non-intervention benchmark, where the short end is market-determined and the relevant lower boundary is zero.

In the present extension, Axiom 4 is better understood as a boundary closure condition rather than as a universal boundary for every policy regime. Monetary-policy intervention changes the short-end closure while leaving Axioms 1–3 intact. The original positivity boundary is therefore the Neumann/free-market closure; a fixed policy rate is the Dirichlet closure; and partial policy anchoring is the Robin closure.

This avoids treating policy intervention as an ad hoc change to the stochastic law. The stochastic-evolution, rolling-invariance, and moment-transport machinery remain the same. What changes is the admissible boundary behaviour at zero remaining maturity. Schematically:

$$\begin{aligned}
\text{Non-intervention closure:} \quad & g_N(0,t,t) = 0, \\
\text{Dirichlet policy closure:} \quad & x^D_{t,t} = x_D, \qquad f_D(x_D,t,t) = g_D(x_D,t,t) = 0, \\
\text{Robin policy closure:} \quad & \left.\frac{\partial x^R_{t,t+\tau}}{\partial \tau}\right|_{\tau=0} = -\chi(x^R_{t,t} - x_D).
\end{aligned}$$

The Dirichlet case should therefore be read as a clean limiting benchmark, not as a literal description of monetary policy. The Robin case is the more realistic intervention regime: policy attempts to moderate short-term interest-rate moves around an objective, reflecting mandates such as price stability and growth, but it does not eliminate all short-end randomness.

3 Modelling conventions

The analysis in this note uses the following modelling conventions, which are maintained throughout the remainder of the paper.

- The repeated-root normalization is fixed as the theoretical coalescing-root benchmark. Empirical amplitude flexibility may be introduced separately through a multiplicative parameter η_D, but $\eta_D = 1$ is the structural benchmark.
- The paper keeps a_N, a_D, and a_R distinct throughout. It makes no common-a comparison and does not assume cross-regime equality of MSNR or total variance.
- The Robin case is restricted to $r = 1$ in this paper. A full general-r Robin condition is a separate extension rather than part of the repeated-root bridge developed here.
- Axioms 1–3 are retained unchanged. Axiom 4 is treated as the boundary-regime closure condition: the original zero-positivity boundary is the non-intervention closure, while Dirichlet and Robin are policy-boundary closures.
- The notation follows the convention stated above: g is the half-volatility coefficient with $\sigma = \sqrt{2}g$, and variance rates are written as $\partial_t \operatorname{Var} = 2g^2$. The loading H is the maturity profile satisfying $g_B = a_B H_B$ along the relevant branch.

4 General Dirichlet construction for $0 < r_D \leq 1$

Let x_D denote the policy boundary imposed at the short end. Under the Dirichlet condition, the short end is fixed:

$$x^D_{t,t} = x_D. \tag{4.1}$$

Use regime-specific parameters

$$a_D, \qquad r_D = \frac{\lambda_D^2}{a_D^2}, \qquad 0 < r_D \le 1. \tag{4.2}$$

The Dirichlet analogue of the IRFR drift-diffusion pair is obtained by shifting the vanishing point of f and g from the original zero boundary to x_D. A convenient shifted coordinate is

$$\tilde{x}_{t,T}^D := x_{t,T} - x_\infty - (x_D - x_\infty)e^{-r_D a_D^2 (T-t)}. \tag{4.3}$$

Then the Dirichlet drift and diffusion are

$$f_D(x,t,T) = a_D^2 \tilde{x}_{t,T}^D, \qquad g_D(x,t,T) = a_D \tilde{x}_{t,T}^D. \tag{4.4}$$

Equivalently, written directly in the original state x,

$$f_D(x,t,T) = a_D^2 \left[x - x_\infty - (x_D - x_\infty)e^{-r_D a_D^2 (T-t)}\right],$$
$$g_D(x,t,T) = a_D \left[x - x_\infty - (x_D - x_\infty)e^{-r_D a_D^2 (T-t)}\right].$$

These satisfy the fixed-boundary conditions

$$f_D(x_D,t,t) = 0, \qquad g_D(x_D,t,t) = 0. \tag{4.5}$$

5 Dirichlet term structure for general r_D

The corresponding Dirichlet curve is the two-loading curve

$$x_{t,t+\tau}^D = x_\infty + (x_D - x_\infty)\left[\frac{1}{1+r_D}e^{-r_D a_D^2 \tau} + \frac{r_D}{1+r_D}e^{-a_D^2 \tau}\right]. \tag{5.1}$$

It satisfies

$$x_{t,t}^D = x_D, \qquad \lim_{\tau\to\infty} x_{t,t+\tau}^D = x_\infty. \tag{5.2}$$

For $0 < r_D < 1$, the two maturity loadings are genuinely distinct. The case $r_D = 1$ is not obtained by simply inserting $r_D = 1$ into this distinct-root formula. When the roots coalesce, the independent boundary mode changes form. This is why the case $r_D = 1$ is developed separately below.

Reconciliation of curve normalizations. The curve above is the representation implied by the shifted-coordinate Dirichlet closure used in the previous section. It is obtained by choosing the Dirichlet equilibrium displacement so that the shifted coordinate vanishes at the policy boundary and the long-maturity limit is x_∞.

An earlier candidate form,

$$x_\infty + (x_D - x_\infty)\left[(1+r)e^{-ra^2\tau} - re^{-a^2\tau}\right], \tag{5.3}$$

uses a different amplitude normalization and a different sign convention for the second loading. Although it also satisfies the endpoint $x_{t,t}^D = x_D$, it is not the representation adopted here because it is not the one generated by the shifted coordinate $\tilde{x}_{t,T}^D$. The paper therefore uses the plus-coefficient form throughout the distinct-root Dirichlet construction. This choice keeps the term structure, stochastic loading, and variance-rate formula aligned with the shifted-coordinate closure.

6 Dirichlet instantaneous variance for general r_D

Substituting the general Dirichlet curve in equation (5.1) into the shifted coordinate in equation (4.3) gives the distinct-root stochastic loading

$$H_D^{(r)}(\tau) = \frac{r_D}{1+r_D}(x_D - x_\infty)\left(e^{-a_D^2\tau} - e^{-r_D a_D^2\tau}\right). \tag{6.1}$$

The substitution is direct. From Section 3, the shifted coordinate is

$$\widetilde{x}_{t,t+\tau}^D = x_{t,t+\tau}^D - x_\infty - (x_D - x_\infty)e^{-r_D a_D^2\tau}. \tag{6.2}$$

Using the Section 4 curve, subtracting the boundary displacement leaves $H_D^{(r)}(\tau)$. Since $g_D = a_D H_D^{(r)}$, the variance-rate identity is

$$\frac{\partial}{\partial t}\operatorname{Var}(x_{t,t+\tau}^D) = 2g_D^2. \tag{6.3}$$

Thus the local instantaneous variance rate associated with the loading in equation (6.1) is

$$\frac{\partial}{\partial t}\operatorname{Var}(x_{t,t+\tau}^D) = 2a_D^2\left(\frac{r_D}{1+r_D}\right)^2(x_D - x_\infty)^2\left(e^{-a_D^2\tau} - e^{-r_D a_D^2\tau}\right)^2. \tag{6.4}$$

This expression vanishes at $\tau = 0$ and as $\tau \to \infty$. It is the correct distinct-root Dirichlet formula for $0 < r_D < 1$. If r_D is set equal to one inside this expression, the two exponentials cancel and the variance rate vanishes identically. That cancellation is a warning sign, not an economic result.

7 The case $r_D = 1$ as a coalescing-root boundary problem

The $r_D = 1$ branch is not a generic point inside the Dirichlet family. It is a coalescing-root case. For $r_D \neq 1$, the two independent exponential modes are

$$e^{-a_D^2\tau}, \qquad e^{-r_D a_D^2\tau}. \tag{7.1}$$

As r_D tends to one, these two modes collapse into the same exponential. The non-trivial second independent boundary mode is obtained by a difference quotient:

$$\frac{e^{-r_D a_D^2\tau} - e^{-a_D^2\tau}}{1 - r_D} \longrightarrow a_D^2\tau e^{-a_D^2\tau}. \tag{7.2}$$

The rate at which the distinct-root loading degenerates can be made explicit. Start from

$$H_D^{(r)}(\tau) = \frac{r_D}{1+r_D}(x_D - x_\infty)\left(e^{-a_D^2\tau} - e^{-r_D a_D^2\tau}\right). \tag{7.3}$$

Let $r_D = 1 + \varepsilon$. Then, to first order in ε,

$$e^{-r_D a_D^2\tau} = e^{-a_D^2\tau}e^{-\varepsilon a_D^2\tau} \approx e^{-a_D^2\tau}\left(1 - \varepsilon a_D^2\tau\right), \tag{7.4}$$

and therefore

$$e^{-a_D^2\tau} - e^{-r_D a_D^2\tau} \approx \varepsilon a_D^2\tau e^{-a_D^2\tau}, \qquad \frac{r_D}{1+r_D} \to \frac{1}{2}. \tag{7.5}$$

Hence

$$H_D^{(r)}(\tau) \approx \varepsilon\,\frac{a_D^2}{2}(x_D - x_\infty)\tau e^{-a_D^2\tau}. \tag{7.6}$$

Thus the distinct-root loading vanishes linearly as $r_D \to 1$, and the corresponding variance rate vanishes quadratically. The non-trivial repeated-root loading is obtained by extracting the Taylor coefficient at the coalescence point.

Equivalently, the repeated-root Dirichlet loading generated by the limit in equation (7.2) can be written as

$$H_D^{(1)}(\tau) = C_D \tau e^{-a_D^2 \tau}, \tag{7.7}$$

with the theoretical repeated-root normalization

$$C_D = \frac{a_D^2}{2}(x_D - x_\infty). \tag{7.8}$$

The normalization in equation (7.8) is fixed in the theoretical benchmark because the paper is deriving the humped loading from a boundary condition, not imposing an empirical amplitude. The sign of C_D follows the convention used in the difference quotient. Reversing the quotient reverses the sign of C_D, but leaves the variance rate and local noise scale unchanged.

For empirical comparisons, the paper may introduce a separate multiplicative amplitude η_D:

$$H_{D,\text{emp}}^{(1)}(\tau) = \eta_D C_D \tau e^{-a_D^2 \tau}. \tag{7.9}$$

The structural benchmark is $\eta_D = 1$. Values not equal to one allow empirical amplitude distortion without weakening the theoretical claim that the shape is generated by the boundary condition.

The corresponding repeated-root variance-rate shape is

$$\frac{\partial}{\partial t} \operatorname{Var}(x_{t,t+\tau}^D) = 2a_D^2 C_D^2 \tau^2 e^{-2a_D^2 \tau}. \tag{7.10}$$

This is the structural source of the volatility loading. It is generated by the fixed short-end boundary as a repeated-root mode; it is not imposed exogenously.

8 Repeated-root benchmark: Neumann versus Dirichlet

Before introducing Robin anchoring, it is useful to put the two limiting $r = 1$ repeated-root benchmarks next to each other. The comparison shows exactly what the boundary condition changes. The Neumann case retains the ordinary exponential stochastic mode at the short end. The Dirichlet case removes that ordinary short-end stochastic mode and replaces it with the repeated-root boundary mode.

Neumann/free boundary. In the $r = 1$ Neumann benchmark, write the equilibrium curve as

$$m_N(\tau) = x_\infty + A_N e^{-a_N^2 \tau}. \tag{8.1}$$

The associated stochastic loading is

$$H_N(\tau) = A_N e^{-a_N^2 \tau}, \tag{8.2}$$

and the instantaneous variance rate is

$$\frac{\partial}{\partial t} \operatorname{Var}(x_{t,t+\tau}^N) = 2a_N^2 A_N^2 e^{-2a_N^2 \tau}. \tag{8.3}$$

The short-end volatility is non-zero whenever A_N is non-zero. This is the free-boundary benchmark: stochastic exposure is allowed to enter immediately at zero maturity.

Dirichlet repeated-root boundary. In the $r = 1$ Dirichlet benchmark, the equilibrium curve lies in the repeated-root linear-exponential family

$$m_D(\tau) = x_\infty + (A_D + B_D\tau)e^{-a_D^2\tau}. \tag{8.4}$$

The pinned short-end condition gives

$$m_D(0) = x_\infty + A_D = x_D. \tag{8.5}$$

The level condition therefore fixes

$$A_D = x_D - x_\infty. \tag{8.6}$$

The coefficient B_D controls the interior curvature of the repeated-root equilibrium curve unless an additional slope or calibration condition is imposed.

The corresponding repeated-root stochastic loading is

$$H_D(\tau) = C_D\tau e^{-a_D^2\tau}, \tag{8.7}$$

with the theoretical coalescing-root normalisation

$$C_D = \frac{a_D^2}{2}(x_D - x_\infty). \tag{8.8}$$

Hence

$$\frac{\partial}{\partial t}\operatorname{Var}(x^D_{t,t+\tau}) = 2a_D^2 C_D^2 \tau^2 e^{-2a_D^2\tau}. \tag{8.9}$$

The Dirichlet boundary therefore forces the local volatility to vanish at $\tau = 0$, while allowing stochastic exposure to appear inside the maturity curve.

The coefficients B_D and C_D play different roles. The coefficient B_D determines the repeated-root component of the deterministic mean curve $m_D(\tau)$, while C_D determines the repeated-root stochastic loading $H_D(\tau)$. The coalescing-root normalization fixes C_D in the structural benchmark. By contrast, B_D requires a separate slope condition, calibration choice, or economic normalization. The two coefficients should therefore not be identified unless an additional modelling restriction is imposed.

Expected values. The KFE first-moment identity gives the same conditional-expectation structure in each benchmark, with its own regime-specific equilibrium curve and drift-diffusion scale. For $B \in \{N, D\}$,

$$\mathbb{E}_t[x^B_{t+s,T}] = m_B(T - t - s) + \left(x^B_{t,T} - m_B(T - t)\right) e^{-a_B^2 s}. \tag{8.10}$$

Thus the distinction between the two benchmarks is not the form of moment transport, but the admissible equilibrium curve and stochastic loading generated by the boundary condition.

Object	Neumann/free $r = 1$	Dirichlet repeated-root $r = 1$
Short-end condition	Free/stochastic	Fixed at x_D
Equilibrium curve	$m_N(\tau) = x_\infty + A_N e^{-a_N^2\tau}$	$m_D(\tau) = x_\infty + (A_D + B_D\tau)e^{-a_D^2\tau}$, $A_D = x_D - x_\infty$
Stochastic loading	$H_N(\tau) = A_N e^{-a_N^2\tau}$	$H_D(\tau) = C_D\tau e^{-a_D^2\tau}$
Short-end volatility	Non-zero if $A_N \neq 0$	Zero at $\tau = 0$
Variance rate	$2a_N^2 A_N^2 e^{-2a_N^2\tau}$	$2a_D^2 C_D^2 \tau^2 e^{-2a_D^2\tau}$
MSNR	$a_N/\sqrt{2}$	$a_D/\sqrt{2}$

This comparison is the core boundary effect in the repeated-root setting: Neumann keeps the ordinary exponential stochastic mode, while Dirichlet replaces it with the repeated-root boundary mode. Robin, developed next, interpolates between these limiting cases by retaining a residual short-end component while also allowing the boundary-induced repeated-root component.

9 Robin repeated-root derivation and Ritchken–Chuang mapping

The repeated-root loading provides the bridge to the more common exogenous humped-volatility specifications in the interest-rate literature. This section develops the Robin case from the repeated-root solution space before mapping it to Ritchken–Chuang-style notation.

A notational distinction is important. The Robin boundary strength is denoted by χ. The exponential decay parameter in the empirical humped-volatility notation is denoted by γ. This avoids using the same symbol for two conceptually different objects.

In the $r = 1$ repeated-root branch, the independent maturity modes are

$$e^{-a_R^2\tau}, \qquad \tau e^{-a_R^2\tau}. \tag{9.1}$$

Therefore any Robin loading or equilibrium curve built in this branch should live in the linear-exponential span generated by these two functions.

The restriction to $r = 1$ in the Robin development is deliberate rather than merely incomplete. It is the minimal repeated-root setting in which the boundary-induced hump appears in linear-exponential form, making the connection to the standard humped-volatility literature transparent. A full general-r Robin theory would be a separate extension, not the object of the present bridge paper.

10 Robin mean curve and boundary condition

Let a_R denote the Robin drift-diffusion scale. A compatible Robin equilibrium or mean curve is

$$m_R(\tau) = x_\infty + (A_R + B_R\tau)e^{-a_R^2\tau}. \tag{10.1}$$

At the short end,

$$m_R(0) = x_\infty + A_R. \tag{10.2}$$

Differentiating gives

$$m_R'(\tau) = \left[B_R - a_R^2(A_R + B_R\tau)\right] e^{-a_R^2\tau}, \tag{10.3}$$

and hence

$$m_R'(0) = B_R - a_R^2 A_R. \tag{10.4}$$

The Robin boundary condition links the short-end slope to the deviation of the short-end curve from the policy boundary x_D:

$$m_R'(0) = -\chi(m_R(0) - x_D). \tag{10.5}$$

Substituting the value and derivative at the boundary gives the coefficient restriction

$$B_R - a_R^2 A_R = -\chi(x_\infty + A_R - x_D). \tag{10.6}$$

Equivalently,

$$B_R = a_R^2 A_R - \chi(x_\infty + A_R - x_D). \tag{10.7}$$

Equation (10.7) also gives a convenient formula for the shape of a Robin IRFR curve at calendar time t. Write the current short-end value as

$$x_{t,t}^R = x_R(t), \tag{10.8}$$

and set

$$A_R(t) := x_R(t) - x_\infty. \tag{10.9}$$

The repeated-root Robin curve is then

$$x_{t,t+\tau}^R = x_\infty + [A_R(t) + B_R(t)\tau]\, e^{-a_R^2\tau}, \tag{10.10}$$

where the Robin boundary condition determines the slope coefficient as

$$B_R(t) = a_R^2(x_R(t) - x_\infty) - \chi(x_R(t) - x_D). \tag{10.11}$$

Equivalently, the curve may be written directly in terms of the current short rate:

$$x_{t,t+\tau}^R = x_\infty + \left[(x_{t,t}^R - x_\infty) + \left\{a_R^2(x_{t,t}^R - x_\infty) - \chi(x_{t,t}^R - x_D)\right\}\tau\right] e^{-a_R^2\tau}. \tag{10.12}$$

Equation (10.12) is the Robin analogue of the repeated-root IRFR curve shape. It starts from the observed short-end value, imposes the mixed boundary restriction at $\tau = 0$, and converges to x_∞ as $\tau \to \infty$.

Illustration: the neutral boundary $x_D = x_\infty/2$. A useful benchmark is to set the policy boundary equal to the $r = 1$ non-intervention short-end equilibrium,

$$x_D = \frac{x_\infty}{2}. \tag{10.13}$$

Then the Robin slope coefficient becomes

$$B_R(t) = a_R^2(x_R(t) - x_\infty) - \chi\left(x_R(t) - \frac{x_\infty}{2}\right). \tag{10.14}$$

The second term is the policy-boundary response. If the short end lies above the neutral boundary, $x_R(t) > x_\infty/2$, the Robin term reduces the initial slope relative to the unconstrained repeated-root component. If the short end lies below the neutral boundary, $x_R(t) < x_\infty/2$, the Robin term increases the initial slope. Under the sign convention used in this paper, this can be described as a loosening response when the curve is above the neutral boundary and a tightening response when the curve is below it. If an empirical implementation uses the opposite sign convention for the IRFR state, these labels should be reversed; the invariant object is the boundary-response term $-\chi(x_R(t) - x_D)$.

This restriction is one equation in two coefficients. Thus the Robin mean curve is a one-parameter family unless an additional normalisation is imposed. In this interpretation, A_R controls the short-end level $m_R(0)$, while χ controls how strongly the initial slope reacts to the deviation of that short-end level from the policy boundary. Empirical or policy calibration may choose A_R directly; the Robin boundary condition then determines B_R.

Thus Robin is a partial-policy boundary. It does not force $m_R(0) = x_D$. Instead, it says that when the short end deviates from the policy boundary, the initial slope of the curve responds in a direction that moderates that deviation. Economically, χ is the policy reaction strength at the short-end boundary. A small χ corresponds to weak anchoring and a boundary closer to the free-market case. A large χ corresponds to strong policy correction: deviations from x_D generate a steeper boundary response. In the formal anchoring limit $\chi \to \infty$, with $m_R(0) \to x_D$, the Robin boundary approaches the Dirichlet fixed-boundary benchmark.

11 Robin stochastic loading and instantaneous variance

The stochastic loading should be built in the same repeated-root function space as the mean curve. Write

$$H_R(\tau) = (C_0 + C_1\tau)e^{-a_R^2\tau}. \tag{11.1}$$

Here C_0 is residual short-end stochastic loading and $C_1\tau$ is the boundary-induced or policy-transition loading. The Dirichlet limit corresponds to $C_0 = 0$; a non-zero C_0 means the short end remains partly stochastic.

Proposition 1 (Boundary-induced linear-exponential volatility mode)**.** *In the repeated-root branch $r = 1$, a Dirichlet or Robin short-end boundary generates a linear-exponential stochastic loading. In the Dirichlet limiting case, the loading is proportional to $\tau e^{-a_D^2\tau}$. In the Robin case, the admissible loading takes the form $(C_0 + C_1\tau)e^{-a_R^2\tau}$.*

Proof sketch. For $r_D \neq 1$, the distinct-root Dirichlet modes are $e^{-a_D^2\tau}$ and $e^{-r_D a_D^2\tau}$. As r_D tends to one, these modes coalesce. The non-trivial second mode is obtained by extracting the first-order Taylor coefficient of their difference, giving $\tau e^{-a_D^2\tau}$. The Robin boundary retains the same repeated-root solution space but allows a residual short-end component in addition to the boundary-induced component. Hence the Robin loading is linear-exponential. This proves that the humped volatility shape is generated by the boundary condition rather than imposed as a primitive volatility function. □

The Dirichlet repeated-root case is recovered from the Robin loading by setting $C_0 = 0$ and $C_1 = C_D$. At the boundary-condition level it is recovered as the strong-anchoring limit $\chi \to \infty$, with $m_R(0) \to x_D$, so that the short end is forced onto the policy boundary. Robin is therefore genuinely an interpolating construction: finite χ permits partial anchoring, while the limiting case recovers the fixed-boundary benchmark.

It is important not to collapse the state-dependent drift into a maturity-only object too early. The structural affine Robin drift should be written as

$$f_R(x, t, T) = a_R^2\,[x - m_R(\tau)] + m_R'(\tau), \qquad \tau = T - t. \tag{11.2}$$

The volatility loading is represented separately by $H_R(\tau)$. Along the relevant mean/loading branch, the local signal and noise share this same H_R.

The diffusion coefficient in the IRFR convention is

$$g_R(\tau) = a_R H_R(\tau), \tag{11.3}$$

and since $\sigma_R = \sqrt{2}\, g_R$, the local volatility is

$$\sigma_R(\tau) = \sqrt{2}\, a_R \left| H_R(\tau) \right|. \tag{11.4}$$

The instantaneous variance rate is therefore

$$\frac{\partial}{\partial t} \operatorname{Var}(x^R_{t,t+\tau}) = 2a_R^2 \left(C_0 + C_1 \tau \right)^2 e^{-2a_R^2 \tau}. \tag{11.5}$$

This is the structural Robin variance form: the loading is mixed first, and the variance is the square of that mixed loading.

The mechanism is therefore simple. The boundary suppresses, or partially suppresses, stochastic loading at $\tau = 0$. Because diffusion still operates through the curve, the stochastic exposure is redistributed away from the short end. The result is a front-end loading when policy is absent, a pure interior hump in the Dirichlet limit, and a mixed front-end-plus-hump profile under Robin anchoring.

11.1 Admissibility and positivity of repeated-root curves

A boundary condition fixes or constrains the behaviour at zero maturity, but it does not by itself guarantee that the entire repeated-root curve remains positive over the maturity domain. For the repeated-root Dirichlet and Robin cases, positivity is therefore an admissibility condition on the coefficients of the linear-exponential curve.

Consider a generic repeated-root mean curve

$$m(\tau) = x_\infty + (A + B\tau) e^{-a^2 \tau}. \tag{11.6}$$

Differentiating with respect to maturity gives

$$m'(\tau) = \Big[B - a^2 (A + B\tau) \Big] e^{-a^2 \tau}. \tag{11.7}$$

Since the exponential term is strictly positive, an interior stationary point satisfies

$$\tau^* = \frac{B - a^2 A}{a^2 B}, \qquad B \neq 0, \tag{11.8}$$

provided $\tau^* > 0$. The stationary point is a maximum when $B > 0$ and a minimum when $B < 0$.

The positivity restriction is only non-trivial when the repeated-root curve has an interior minimum. Thus suppose

$$B < 0, \qquad \tau_{\min} = \frac{B - a^2 A}{a^2 B} > 0. \tag{11.9}$$

At the minimum,

$$A + B\tau_{\min} = \frac{B}{a^2}. \tag{11.10}$$

Therefore

$$m(\tau_{\min}) = x_\infty + \frac{B}{a^2} \exp\left(-1 + \frac{a^2 A}{B} \right). \tag{11.11}$$

The repeated-root curve is admissible only if the interior minimum remains positive:

$$x_\infty + \frac{B}{a^2} \exp\left(-1 + \frac{a^2 A}{B} \right) > 0. \tag{11.12}$$

Equivalently, for $B < 0$,

$$\frac{|B|}{a^2} \exp\left(-1 + \frac{a^2 A}{B}\right) < x_\infty. \tag{11.13}$$

This condition formalises the admissibility restriction used in the stylised example below: the Robin boundary may generate an interior trough, but the parameters must keep that trough above zero.

For the stylised Robin sign convention used in the figures below,

$$A_j = x_j - x_\infty, \qquad B_j = a_R^2(x_j - x_\infty) + \chi(x_j - x_D). \tag{11.14}$$

The below-equilibrium case has $B_- < 0$ and hence an interior minimum. The illustrative parameter choice in equation (11.16) gives $\tau_{\min} \approx 5.1$ years and $m(\tau_{\min}) \approx 0.46\%$, so the plotted curve satisfies the positivity condition in equation (11.12).

11.2 Stylised 80/20 Robin curve and volatility illustrations

The neutral-boundary convention in equation (10.13) can be illustrated by choosing three short-end states

$$x_- < x_D, \qquad x_0 = x_D, \qquad x_+ > x_D. \tag{11.15}$$

Here x_- is interpreted as the below-equilibrium or tightening case, x_0 as the neutral case, and x_+ as the above-equilibrium or loosening case. To make the illustration concrete, the figures use

$$x_\infty = 4\%, \qquad x_D = \frac{x_\infty}{2} = 2\%, \qquad x_- = 1.5\%, \qquad x_0 = 2.0\%, \qquad x_+ = 2.5\%, \qquad a_R = 0.35, \qquad \chi = 1. \tag{11.16}$$

For the stylised yield-curve plot, the Robin repeated-root curve is written as

$$x_j^R(\tau) = x_\infty + (A_j + B_j\tau)e^{-a_R^2\tau}, \qquad j \in \{-, 0, +\}, \tag{11.17}$$

where

$$A_j := x_j - x_\infty, \qquad B_j := a_R^2(x_j - x_\infty) + s\chi(x_j - x_D). \tag{11.18}$$

The sign parameter s records the convention used to label movements around the policy boundary. The figures use $s = +1$, so that an above-equilibrium short end generates an upward boundary response and can produce an interior maximum in the IRFR curve. This sign convention is illustrative; it is the direction of the boundary-response term, rather than the label attached to it, that carries the mathematical content.

With the parameters in equation (11.16), the below-equilibrium curve remains positive and has an interior minimum of approximately 0.46% at about 5.1 years. The above-equilibrium curve has an interior maximum of approximately 4.53% at about 12.9 years. The neutral curve is monotone in the displayed maturity range. These turning points follow from differentiating equation (11.17): if $B_j \neq 0$, the stationary maturity is

$$\tau_j^* = \frac{B_j - a_R^2 A_j}{a_R^2 B_j}, \tag{11.19}$$

provided $\tau_j^* > 0$. The stationary point is a maximum when $B_j > 0$ and a minimum when $B_j < 0$.

For the matching stylised volatility curves, the example uses an 80% Neumann and 20% Dirichlet mixture of the repeated-root loadings. Let

$$H_j^{80/20}(\tau) = \left[0.8C_{N,j} + 0.2C_D\tau\right] e^{-a_R^2\tau}, \tag{11.20}$$

where $C_{N,j}$ is the residual market, or Neumann, loading in state j and $C_D\tau$ is the boundary-induced repeated-root, or Dirichlet, loading. The associated illustrative volatility is

$$\sigma_j^{80/20}(\tau) = \sqrt{2}\, a_R \left| H_j^{80/20}(\tau) \right| . \tag{11.21}$$

This is a plotting convention rather than a calibration restriction. It makes explicit the interpretation of Robin as a partial-boundary case: most of the stochastic loading remains market-like, but a smaller repeated-root component generates an interior hump.

Figures 1 and 2 show the resulting illustrative IRFR and volatility curves. The purpose of the figures is not empirical calibration. Rather, they show how the Robin boundary condition can be read as a partial policy response around the neutral fixed point $x_D = x_\infty/2$, and how an 80/20 market-policy mixture preserves residual short-end volatility while retaining a boundary-induced interior hump.

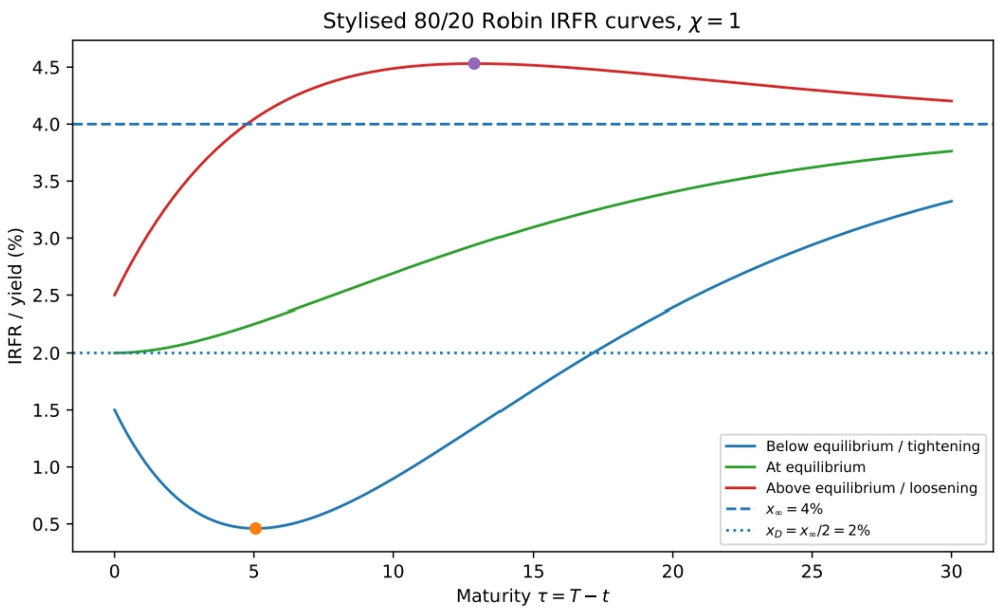

Figure 1: Stylised Robin IRFR curves for a below-equilibrium, neutral, and above-equilibrium short-end state. The figure uses the illustrative fixed point $x_D = x_\infty/2$, boundary strength $\chi = 1$, and the sign convention $s = +1$ in equation (11.18).

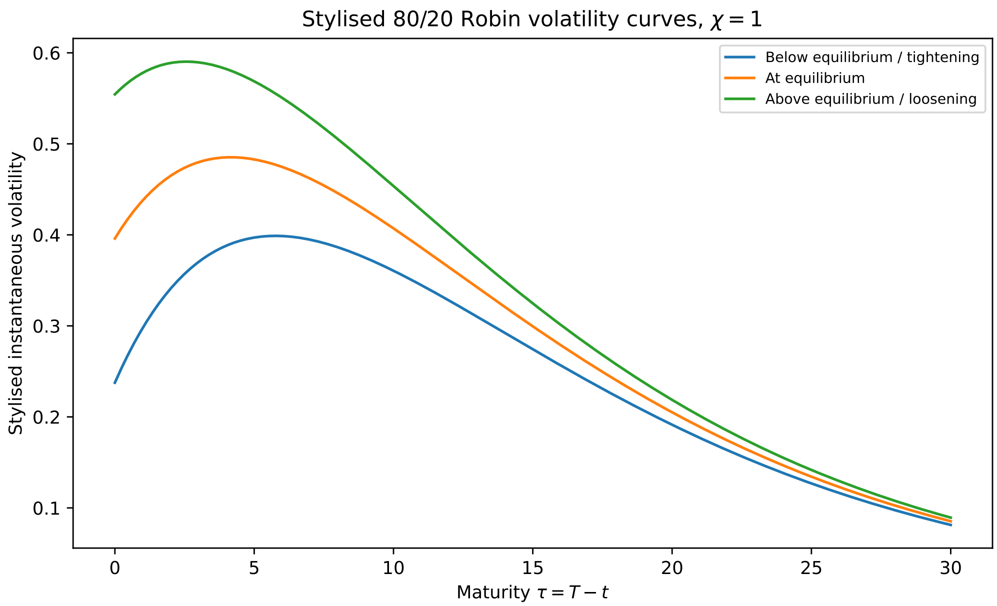

Figure 2: Stylised 80% Neumann and 20% Dirichlet Robin instantaneous volatility curves corresponding to the three short-end states in Figure 1. The curves use equations (11.20)–(11.21).

12 Mapping to Ritchken–Chuang-style humped volatility

The humped-volatility literature typically begins by specifying a maturity-dependent volatility function. Ritchken and Chuang introduce stationary humped forward-rate volatility in an HJM setting. Moraleda and Vorst and Mercurio and Moraleda develop tractable humped-volatility models for interest-rate claims, while Falini documents cap-pricing benefits from humped HJM volatility specifications. The present paper differs from this literature by deriving a linear-exponential humped loading from a boundary condition. The mapping below is therefore not only a change of notation; it identifies the reduced-form humped-volatility parameters with the coefficients of a repeated-root Robin boundary structure.

A common reduced-form humped-volatility notation can be written as

$$\sigma_{RC}(\tau) = (\alpha + \beta\tau)e^{-\gamma\tau}. \tag{12.1}$$

Using the Robin loading in equation (11.1) gives

$$\sigma_R(\tau) = \sqrt{2}a_R(C_0 + C_1\tau)e^{-a_R^2\tau}, \tag{12.2}$$

whenever $C_0 + C_1\tau$ is non-negative on the maturity range. The mapping is then

$$\alpha = \sqrt{2}a_R C_0, \qquad \beta = \sqrt{2}a_R C_1, \qquad \gamma = a_R^2. \tag{12.3}$$

For the pure Dirichlet repeated-root case, $C_0 = 0$ and $C_1 = C_D$, so the loading reduces to

$$\sigma_D(\tau) = \sqrt{2}a_D C_D \tau e^{-a_D^2\tau}, \tag{12.4}$$

and therefore

$$\alpha = 0, \qquad \beta = \sqrt{2}a_D C_D, \qquad \gamma = a_D^2. \tag{12.5}$$

The cleanest empirical comparison is obtained under the following restrictions:

- $\gamma > 0$, so the exponential decay scale is positive.
- $\alpha \geq 0$ and $\beta \geq 0$, so the Robin loading is non-negative over all $\tau \geq 0$.
- For a genuine interior hump with $\alpha > 0$, require $\beta > \gamma\alpha$. The volatility peak is

$$\tau^* = \frac{\beta - \gamma\alpha}{\gamma\beta}. \tag{12.6}$$

- For the pure Dirichlet repeated-root case $\alpha = 0$, the peak is

$$\tau^* = \frac{1}{\gamma} = \frac{1}{a_D^2}. \tag{12.7}$$

This is the central empirical bridge. The Ritchken–Chuang-style formulation specifies a humped volatility function directly. The boundary-condition interpretation derives the same linear-exponential loading from the repeated-root Robin or Dirichlet boundary structure.

13 Expected value under the Robin repeated-root branch: KFE derivation

The Robin expected-value formula follows from the KFE first-moment identity, not merely from analogy with the $r = 1$ transport formula.

Let

$$\tau = T - t, \tag{13.1}$$

and let the Robin equilibrium curve be

$$m_R(\tau) = x_\infty + (A_R + B_R\tau)e^{-a_R^2\tau}. \tag{13.2}$$

Choose the Robin drift in the structural affine form

$$f_R(x, t, T) = a_R^2\,[x - m_R(\tau)] + m_R'(\tau). \tag{13.3}$$

The KFE first-moment identity gives

$$\frac{\partial}{\partial t}\mathbb{E}[x_{t,T}^R] = -\mathbb{E}[f_R(x_{t,T}^R, t, T)]. \tag{13.4}$$

Taking expectations in the affine drift gives

$$\frac{\partial}{\partial t}\mathbb{E}[x_{t,T}^R] = -a_R^2\left(\mathbb{E}[x_{t,T}^R] - m_R(\tau)\right) - m_R'(\tau). \tag{13.5}$$

Because $\tau = T - t$, the equilibrium branch satisfies

$$\frac{\partial}{\partial t}m_R(\tau) = -m_R'(\tau). \tag{13.6}$$

Define the deviation

$$z(t) := \mathbb{E}[x_{t,T}^R] - m_R(T - t). \tag{13.7}$$

Then

$$\frac{\mathrm{d}z(t)}{\mathrm{d}t} = -a_R^2 z(t). \tag{13.8}$$

Solving from t_0 to t gives

$$z(t) = z(t_0)e^{-a_R^2(t-t_0)}. \tag{13.9}$$

Therefore the conditional expectation is

$$\mathbb{E}_{t_0}[x_{t,T}^R] = m_R(T - t) + \left(x_{t_0,T}^R - m_R(T - t_0)\right)e^{-a_R^2(t-t_0)}. \tag{13.10}$$

This is the rigorous Robin expectation formula. It follows directly from the KFE moment identity and the affine Robin drift, with the repeated-root equilibrium curve $m_R(\tau)$.

14 MSNR, total return, and total variance under Robin

Within the Robin regime, the local signal and local noise share the same loading H_R. The signal scale is

$$S_R(\tau) = a_R^2 H_R(\tau), \tag{14.1}$$

while the local volatility scale is

$$\sqrt{v_R(\tau)} = \sqrt{2}a_R|H_R(\tau)|. \tag{14.2}$$

Thus, where $H_R(\tau)$ is non-zero,

$$\mathrm{MSNR}_R(\tau) = \frac{|S_R(\tau)|}{\sqrt{v_R(\tau)}} = \frac{a_R}{\sqrt{2}}. \tag{14.3}$$

The MSNR is maturity-independent within the Robin regime. However, no equality between a_N, a_D, and a_R is assumed. Cross-regime MSNR comparisons require an explicit additional restriction, which this paper does not impose.

Because the IRFR covariance-rate kernel is rank one, the variance of an integrated maturity object is the square of the integrated stochastic loading. One should integrate volatility loadings first and then square the result, rather than integrating pointwise variances:

$$\mathcal{V}_R^{\mathrm{tot}} = 2a_R^2 \left(\int_0^\infty |H_R(\tau)| \,\mathrm{d}\tau \right)^2. \tag{14.4}$$

In the Robin repeated-root case,

$$\int_0^\infty H_R(\tau)\,\mathrm{d}\tau = \frac{C_0}{a_R^2} + \frac{C_1}{a_R^4}, \tag{14.5}$$

provided a consistent sign convention is used for H_R. If H_R has a consistent sign over the relevant maturity region, then

$$\mathcal{V}_R^{\mathrm{tot}} = 2a_R^2 \left(\frac{C_0}{a_R^2} + \frac{C_1}{a_R^4} \right)^2. \tag{14.6}$$

The corresponding total instantaneous expected return is

$$\mathcal{R}_R^{\mathrm{tot}} = \int_0^\infty a_R^2 H_R(\tau)\,\mathrm{d}\tau = C_0 + \frac{C_1}{a_R^2}. \tag{14.7}$$

Consequently, within a fixed Robin regime,

$$\frac{\mathcal{R}_R^{\mathrm{tot}}}{\sqrt{\mathcal{V}_R^{\mathrm{tot}}}} = \frac{a_R}{\sqrt{2}}, \tag{14.8}$$

again assuming a consistent sign convention for the loading. This is an internal Robin-regime statement, not a claim of equality across Neumann, Dirichlet, and Robin regimes.

15 Conclusion

This paper interprets monetary-policy intervention as a boundary condition on the IRFR curve. The first three IRFR axioms - stochastic evolution, rolling-maturity invariance, and equilibrium moment transport - are retained. What changes across policy regimes is the short-end boundary closure.

The general Dirichlet construction shows how a fixed policy boundary changes the admissible maturity loadings of the forward-rate system. For $0 < r_D < 1$, the Dirichlet branch has two distinct exponential modes. The branch $r_D = 1$ is different: it is a coalescing-root case in which the non-trivial boundary-induced mode is obtained by extracting the repeated-root limit. This produces the linear-exponential loading

$$\tau e^{-a_D^2 \tau}. \tag{15.1}$$

The repeated-root loading gives a structural interpretation of humped volatility forms that are usually specified exogenously. In this interpretation, the volatility hump is not a primitive modelling choice. It is the maturity-domain shadow of a fixed or partially fixed short-end boundary.

The Robin construction then provides the realistic partial-anchoring case. Its mean curve and stochastic loading both live in the repeated-root span generated by $e^{-a_R^2 \tau}$ and $\tau e^{-a_R^2 \tau}$. The Robin boundary leaves some residual short-end stochasticity while allowing policy-induced displacement of stochastic exposure into the interior of the curve. This gives a direct mapping to Ritchken–Chuang-style linear-exponential humped volatility, while preserving the IRFR discipline that a_N, a_D, and a_R remain boundary-regime-specific drift-diffusion scales.

The main message is therefore that monetary policy can be modelled as a boundary operator on the IRFR system. Boundary conditions do not merely alter the short end locally; they determine the admissible shape of volatility across the maturity curve.

References and source connection

This draft is intended as a companion extension to Oza [1], *An Axiomatically Consistent Single-Factor Term Structure Model with Admissible Humped Volatility*, posted as an SSRN working paper in May 2026. The key source results used here are the general r-branch, the affine drift-diffusion form, the instantaneous variance-rate formula, the constant within-regime MSNR result, and the rank-one covariance aggregation logic for integrated maturity objects. The Ritchken–Chuang connection is developed as a literature bridge: the boundary-condition framework derives a humped loading structurally, whereas the earlier literature commonly specifies humped forward-rate volatility functions directly.

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